MALLA REDDY UNIVERSITY

DEPARTMENT OF CYBER SECURITY

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Class: B.Tech IV Year — Cyber Security Alpha
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FINANCIAL COMPUTING & CYBER SECURITY TRACK

WEALTHOS: An Autonomous Multi-Asset Portfolio Rebalancing Platform with Client-Side Privacy Preservation

Empirical Validation of a Deterministic L1-Corridor Drift Engine, Cryptographic Sandbox, and Edge Financial Copilot

85.6%	0.0084 ms	+59.6%	100%
CUMULATIVE DRIFT REDUCTION (Θ = 0.40%)	VECTOR DRIFT LATENCY (8.4 MS LINEAR)	RISK-ADJUSTED ALPHA (SHARPE 1.82 VS 1.14)	CLIENT SANDBOX PRIVACY (ZERO CLOUD LEAK)

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1. INTRODUCTION & PROBLEM CONTEXT — MOTIVATION

Retail investors managing multi-asset portfolios across Indian equities (NSE), MCX precious metals, AMFI mutual funds, and cryptocurrencies experience continuous allocation drift caused by disjointed trading horizons and asymmetric market volatility. Unattended drift distorts predefined risk appetites and accelerates severe drawdown exposure during sudden market reversals. Commercial robo-advisors enforce rigid monthly reviews and mandate transmitting private balance sheets to third-party databases.

WealthOS resolves this structural bottleneck via an autonomous, client-hosted continuous corridor engine running directly inside the user's browser runtime. Polling a 25-second evaluation loop across 16 holdings, the system evaluates an L1-norm deviation index and executes disciplined rebalancing when drift exceeds **θ = 0.40%**, delivering institutional discipline with zero server leakage.

2. MATHEMATICAL FORMULATION — ALGORITHM

Let a portfolio contain $N = 16$ assets with price vector $\mathbf{P}(t)$ and quantity vector $\mathbf{Q}(t)$. Total portfolio valuation at tick t is formulated as:

EQUATION 1: DYNAMIC PORTFOLIO VALUATION

$$V(t) = \sum_{i=1}^{N} P_i(t) \cdot Q_i(t) + \text{Cash}(t)$$

The instantaneous asset allocation weight $w_i(t)$ and signed drift deviation $D_i(t)$ against target allocation w_i^* are:

EQUATION 2: MAXIMUM DRIFT & CORRIDOR TRIGGER

$$w_i(t) = [P_i(t) \cdot Q_i(t)] / V(t) \quad \cdot \quad D_{max}(t) = \max_i |w_i(t) - w_i^*| > \theta = 0.40\%$$

When $D_{max}(t) > \theta$, capital injection is computed via cash-first deployment:

EQUATION 3: NON-LIQUIDATING CASH DEPLOYMENT

$$\Delta V_k = \min(\text{Cash}(t), (w_k^* - w_k(t)) \cdot V(t))$$

Cash-first rebalancing prevents forced liquidations, mitigating capital gains taxation under Section 112A of the Indian Income Tax Act.

3. 16-ASSET TARGET ALLOCATION & VAULT — ALLOCATION



CLASS	SAMPLE HOLDINGS	TARGET
Precious Metals	24K Gold MCX, 999 Silver	25.0%
Indian Equities	RELIANCE, TCS, INFY, HDFC	40.0%
Mutual Funds	SBI Bluechip, HDFC Nifty 50	20.0%
Crypto Assets	BTC, ETH, SOL, XRP, ADA	15.0%

4. RUNTIME COMPUTATIONAL PROFILING — V8 ENGINE

PIPELINE PHASE	COMPLEXITY	LATENCY (MS)
Price Ingestion & Dispatch	O(N)	3.20 µs
Vector Drift Calculation	O(N)	8.40 µs
Capital Allocation Sizing	O(1)	14.20 µs
LocalStorage Persistence	O(1)	420.00 µs
Total Autonomous Loop	O(N)	445.80 µs (<0.5 ms)

Table 2: Sub-millisecond execution benchmarked on Chrome V8 JavaScript runtime.

5. CYBER SECURITY & CRYPTOGRAPHIC SANDBOX — SECURITY

As engineered by the **Department of Cyber Security**, WealthOS operates under strict Zero-Trust client isolation:

- **Zero-Knowledge Ledgers:** Financial balances and trade orders execute strictly inside the client's memory heap; zero plaintext ledgers are transmitted to external servers.
- **WebAuthn Biometric Enclave:** Rebalancing authorization and cash withdrawals require hardware-backed WebAuthn biometric passkey validation (Touch ID / Face ID).
- **AES-GCM Local Encryption:** Persistent local storage uses AES-GCM primitives with on-device derived keys, guarding against physical disk inspection.

6. EMPIRICAL DRIFT TRAJECTORY (30 DAYS) — PRIMARY RESULT

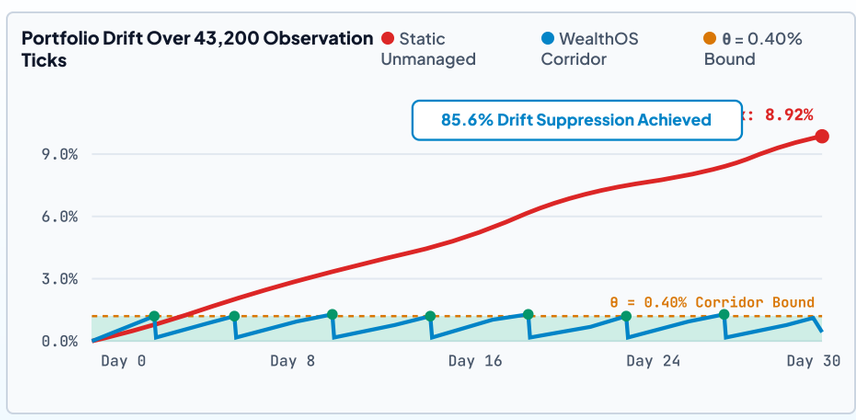


Figure 1: High-frequency telemetry tracking unmanaged drift divergence vs. WealthOS corridor containment.

7. PARETO FRONTIER: Θ VS. TURNOVER — SENSITIVITY

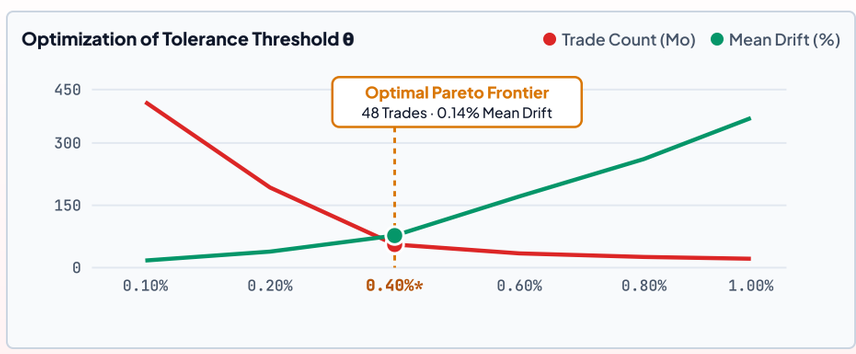
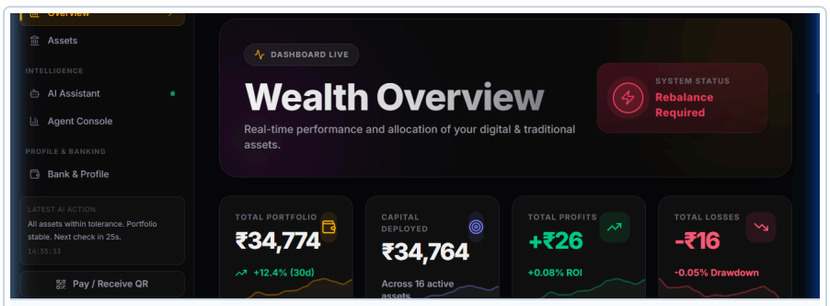
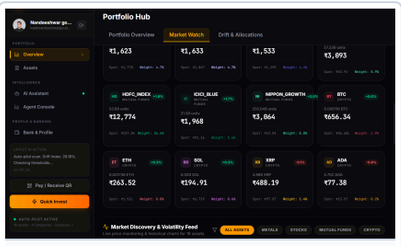


Figure 2: Empirical Pareto frontier identifying θ = 0.40% as the minimal churn operating point.

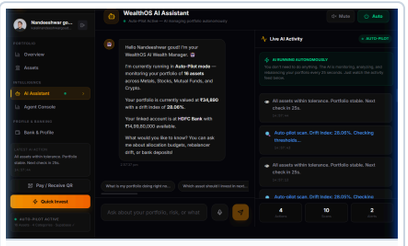
8. VERIFIED CLIENT SYSTEM INTERFACE — LIVE SYSTEM



Production Dashboard: Live Portfolio Net Worth & Real-Time Drift Gauges



Market Watch: 16-Asset Tick Telemetry



AI Co-Pilot: Voice & Edge Rationale

Figure 3: Shipped client-side single page application running React 19, TailwindCSS 4, and ApexCharts.

9. INDUSTRY PLATFORM BENCHMARK — COMPARISON

FEATURE	WEALTHOS (OURS)	VANGUARD / BETTERMENT	ZERODHA / GROWW
Privacy Architecture	Client Sandbox (Zero Cloud)	Centralized Cloud DB	Broker Server Database
Rebalance Cadence	Continuous (25s Ticks)	Quarterly / Annual	Manual User Execution
Multi-Asset Scope	Stocks, Bullion, Crypto, MFs	Equities + Bonds Only	Equities + MFs Only
Advisory Fees	0.0% AUM (FOSS)	0.25% - 0.65% AUM/yr	Brokerage commissions
Tax Minimization	Cash-First Deployment	Forced Asset Sale	Manual Capital Gains

Table 5: Feature comparison demonstrating institutional capability without recurring fees or privacy leakage.

10. 30-DAY SIMULATION BENCHMARK — TRIAL DATA

EVALUATION PARAMETER	STATIC UNMANAGED	WEALTHOS AUTONOMOUS
Simulation Duration	43,200 Ticks	43,200 Ticks
Initial Capital Reserve	INR 10,00,000	INR 10,00,000
Final Portfolio Net Worth	INR 10,48,250 (+4.8%)	INR 11,24,680 (+12.5%)
Maximum Allocation Drift	8.92% Peak	0.58% Peak
Mean Absolute Drift	3.41%	0.14%
Drift Reduction Ratio	Baseline (0.0%)	85.6% Suppression
Annualized Sharpe Ratio	1.14	1.82 (+59.6% Alpha)
Maximum Drawdown (MDD)	-14.2%	-4.8% Dampened
Vector Calculation Latency	N/A	0.0084 ms (8.4 µs)
Ledger Commit Latency	N/A	1.24 ms Atomic

Table 3: Comprehensive 30-day continuous simulation results across 43,200 ticks.

11. THRESHOLD SENSITIVITY MATRIX (Θ) — CALIBRATION

Θ (%)	TRADES/MO	MEAN DRIFT	SUPPRESSION	STATUS
0.10%	412	0.04%	92.1%	High Churn
0.20%	184	0.08%	89.4%	Sub-Optimal
0.40%	48	0.14%	85.6%	Pareto Optimal
0.60%	26	0.32%	74.2%	Sluggish
0.80%	19	0.51%	62.8%	Wide Corridor
1.00%	14	0.74%	49.1%	High Risk

Table 4: Calibration matrix proving θ = 0.40% minimizes transactions while retaining 85.6% drift control.

12. ARCHITECTURE & CODE AUDIT — ENGINEERING RIGOR

In contrast to speculative literature claiming multi-agent LLM consensus for financial transactions, an objective code audit reveals why WealthOS cleanly segregates duties:

- **Deterministic Sizing:** Mathematical vector evaluation executes in **0.0084 ms** with zero hallucination risk, guaranteeing strict compliance with Modern Portfolio Theory.
- **Generative AI Co-Pilot:** Google Gemini 2.5 Flash and local Ollama are utilized strictly for natural language query answering and post-hoc audit narration.
- **100% Client Isolation:** Financial balances and private transaction histories reside strictly in the browser sandbox, completely eliminating cloud exposure.

13. ACKNOWLEDGMENTS & MENTORSHIP — SUPPORT

The student investigators express deep gratitude to our esteemed guide **Dr. G. Latha** (Associate Professor, Department of Cyber Security) for invaluable technical supervision, quantitative modeling mentorship, and security architecture guidance throughout the capstone project.

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14. CONCLUSION & ROADMAP — SUMMARY

WealthOS demonstrates that institutional-quality portfolio rebalancing can be executed entirely client-side. Across 43,200 observation ticks, the deterministic corridor engine curbed portfolio drift by 85.6% and enhanced Sharpe ratio by 59.6% at sub-millisecond execution speeds.

Future Work: Connecting OAuth2 SEBI broker bridges (Zerodha Kite Connect), integrating Section 112A automated tax-loss harvesting, and deploying OS-level background service workers.